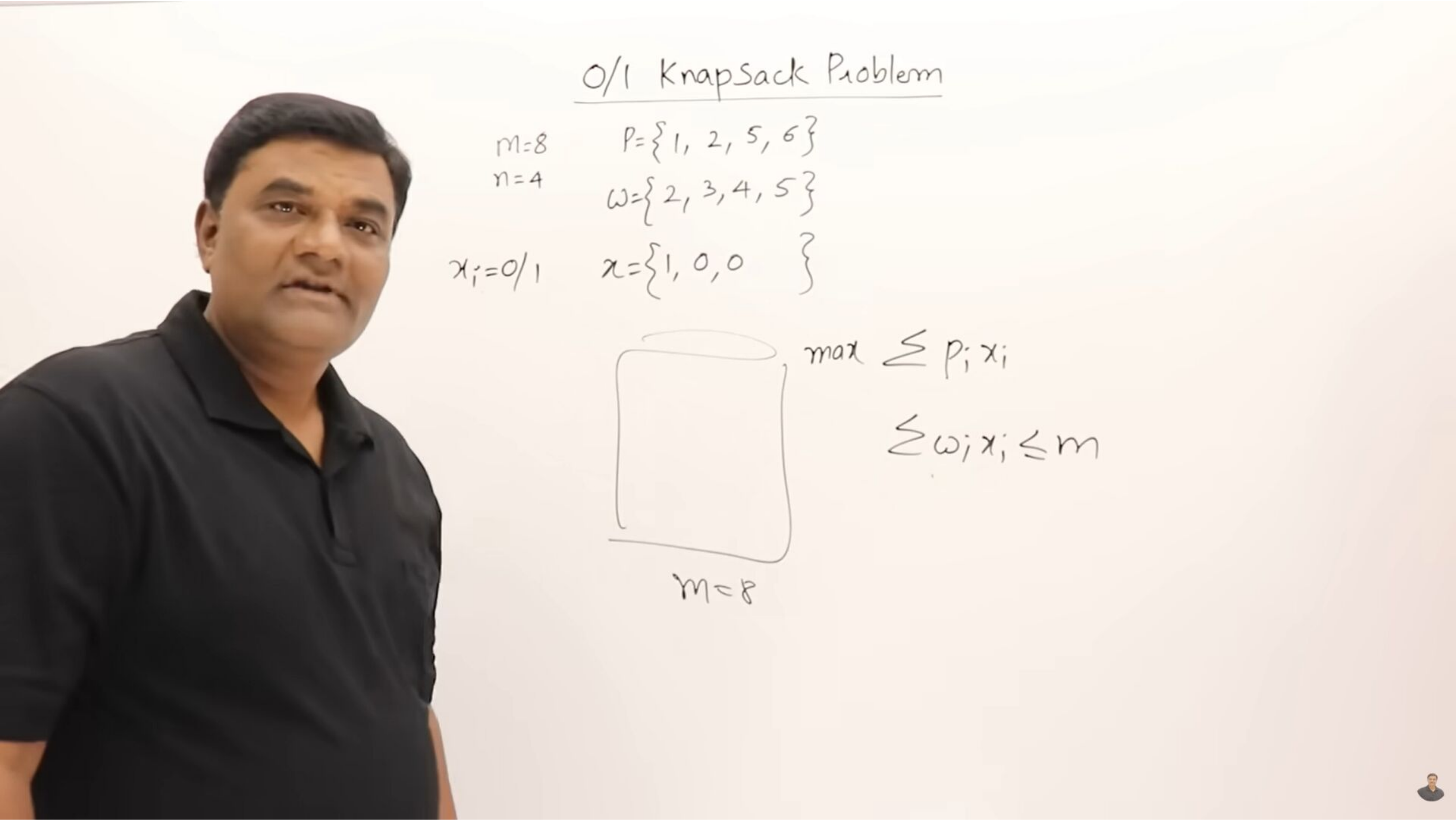

0/1 Knapsack Problem
$m=8$
$n=4$
$P=\{1, 2, 5, 6\}$
$w=\{2, 3, 4, 5\}$
$x_i=0/1$
$x=\{1, 0, 0 \}$
$\max \sum p_i x_i$
$\sum w_i x_i \leq m$
$m=8$

0/1 Knapsack Problem

$m = 8$ $\quad P = \{1, 2, 5, 6\}$

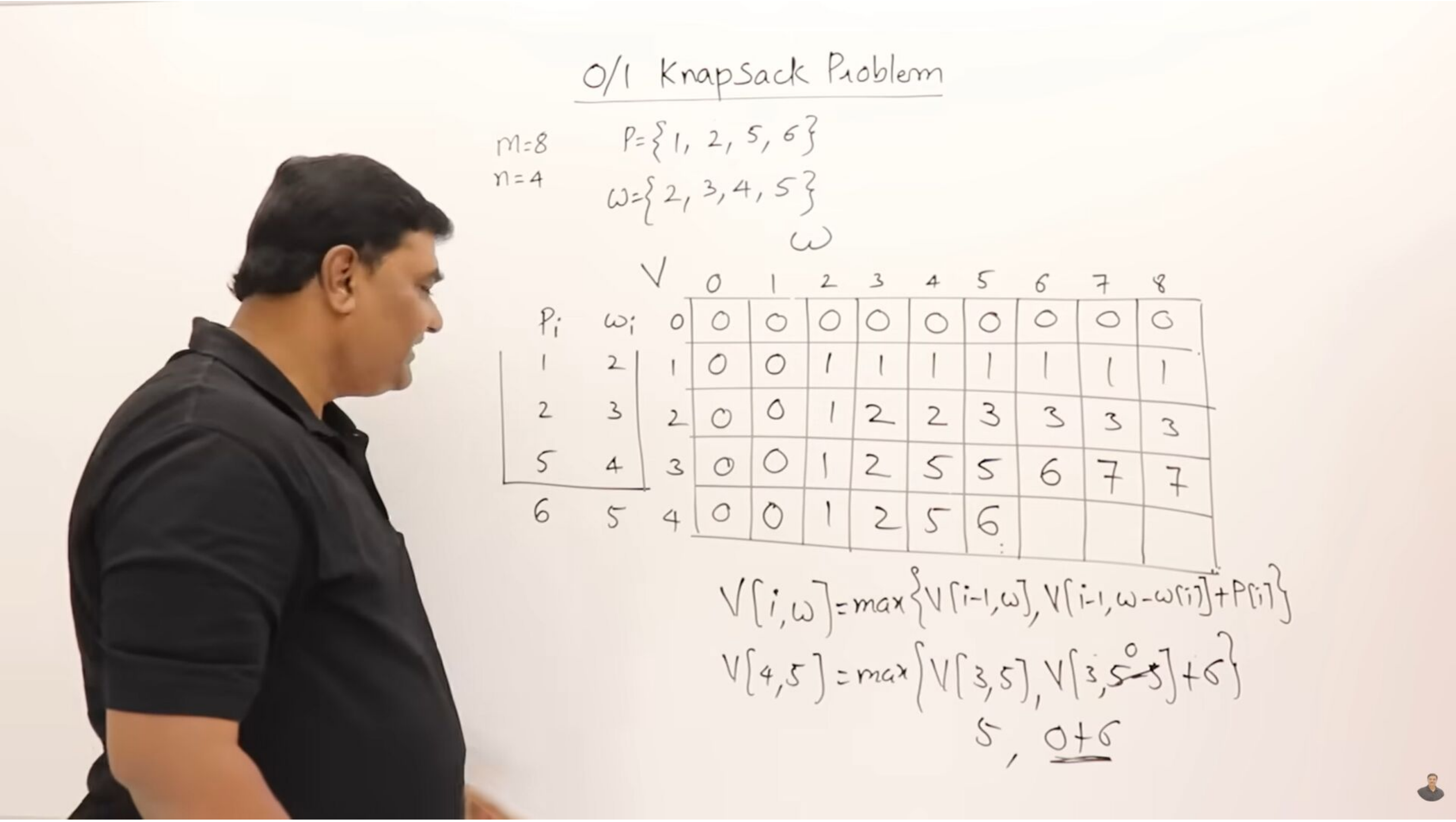
$n = 4$ $\quad w = \{2, 3, 4, 5\}$

P_i	w_i	V	0	1	2	3	4	5	6	7	8
		0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5				

$$V[i, w] = \max\{V[i-1, w],\ V[i-1, w - w[i]] + P[i]\}$$

$$V[4, 1] = \max\{V[3, 1],\ V[3, 1-5] + 6\}$$

$$V[3, -4]$$


0/1 Knapsack Problem
m=8
n=4
P={1, 2, 5, 6}
w={2, 3, 4, 5}
w
V 0 1 2 3 4 5 6 7 8
Pi wi
0 0 0 0 0 0 0 0 0 0
1 2 1 0 0 1 1 1 1 1 1 1
2 3 2 0 0 1 2 2 3 3 3 3
5 4 3 0 0 1 2 5 5 6 7 7
6 5 4 0 0 1 2 5 6
V[i,w]=max{V[i-1,w], V[i-1,w-w[i]]+P[i]}
V[4,5]=max{V[3,5], V[3,5-5]+6}
5, 0+6

0/1 Knapsack Problem

$m=8$ $\quad P=\{1, 2, 5, 6\}$

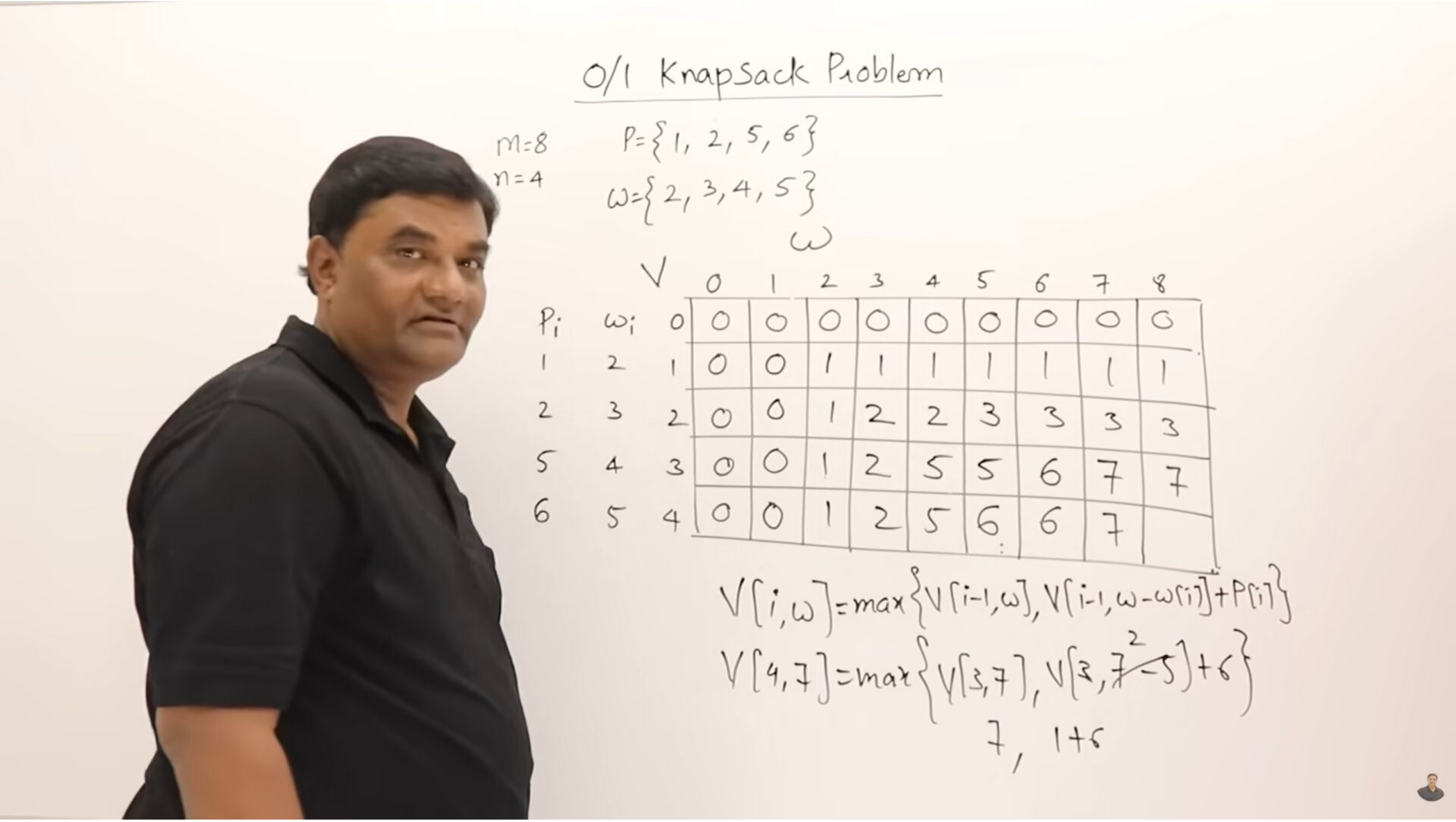
$n=4$ $\quad w=\{2, 3, 4, 5\}$

P_i	w_i	V \ w	0	1	2	3	4	5	6	7	8
		0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6	6		

$$V[i,w] = \max\{V[i-1,w],\ V[i-1, w-w[i]] + P[i]\}$$

$$V[4,6] = \max\{V[3,6],\ V[3,\cancel{6-5}\,1] + 6\}$$

$$6,\ 0+6$$


0/1 Knapsack Problem
m=8
n=4
P={1, 2, 5, 6}
w={2, 3, 4, 5}
w
V 0 1 2 3 4 5 6 7 8
Pi wi 0 0 0 0 0 0 0 0 0 0
1 2 1 0 0 1 1 1 1 1 1 1
2 3 2 0 0 1 2 2 3 3 3 3
5 4 3 0 0 1 2 5 5 6 7 7
6 5 4 0 0 1 2 5 6 6 7
V[i,w]=max{V[i-1,w], V[i-1,w-w[i]]+P[i]}
V[4,7]=max{V[3,7], V[3,7-5]+6}
2
7, 1+6

0/1 Knapsack Problem

$m=8$ $\quad P=\{1, 2, 5, 6\}$

$n=4$ $\quad w=\{2, 3, 4, 5\}$

P_i	w_i	V	0	1	2	3	4	5	6	7	8
		0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6	6	7	8

$$V[i,w] = \max\{V[i-1,w],\ V[i-1, w-w[i]] + P[i]\}$$

$$V[4,8] = \max\{V[3,8],\ V[3, \overset{3}{\cancel{8-5}}] + 6\}$$

$$\underline{7},\ \underline{2+6}$$

0/1 Knapsack Problem

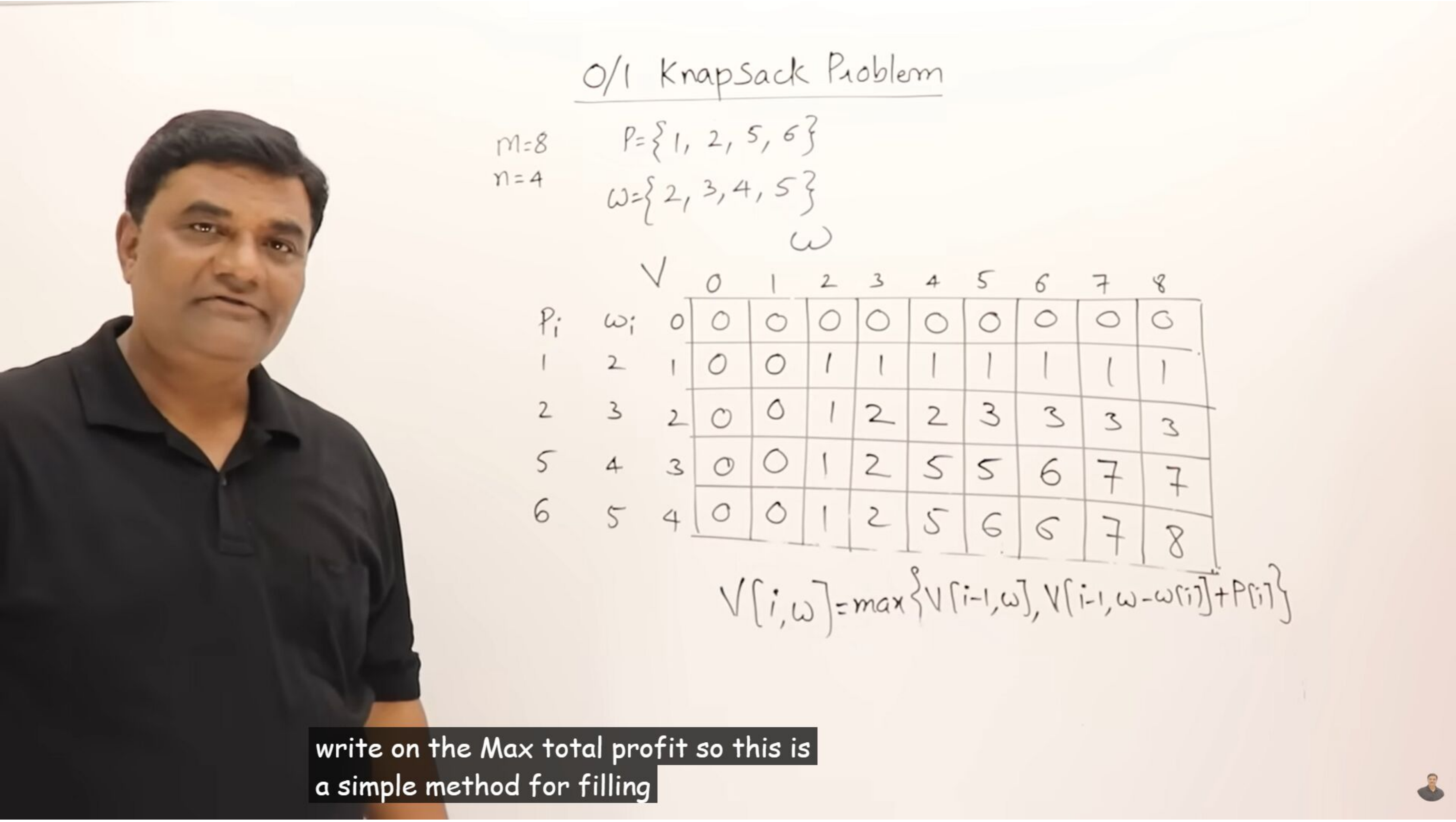
$m=8$ $\quad P=\{1, 2, 5, 6\}$

$n=4$ $\quad w=\{2, 3, 4, 5\}$

P_i	w_i	V	0	1	2	3	4	5	6	7	8
		0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6			

$$V[i,w] = \max\{V[i-1,w],\ V[i-1,w-w[i]]+P[i]\}$$

cells before that same value as the
previous row


0/1 Knapsack Problem
m=8
n=4
P={1, 2, 5, 6}
ω={2, 3, 4, 5}
ω
V 0 1 2 3 4 5 6 7 8
Pi ωi 0 0 0 0 0 0 0 0 0 0
1 2 1 0 0 1 1 1 1 1 1 1
2 3 2 0 0 1 2 2 3 3 3 3
5 4 3 0 0 1 2 5 5 6 7 7
6 5 4 0 0 1 2 5 6 6 7 8
V[i,ω]=max{V[i-1,ω], V[i-1,ω-ω[i]]+P[i]}
write on the Max total profit so this is
a simple method for filling

0/1 Knapsack Problem

$m=8$ $P=\{1, 2, 5, 6\}$

$n=4$ $w=\{2, 3, 4, 5\}$

P_i	w_i	V \ w	0	1	2	3	4	5	6	7	8
		0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6	6	7	8

$\{x_1\ x_2\ x_3\ x_4\}$

$\{0\ 1\ 0\ 1\}$

$8-6=2$

$2-2=0$

is this one so total profit will be
eight so this is the maximum profit